Margeaux Corrigan

Phone: +1 (949) 404-9328 | Margeaux_corrigan@brown.edu | LinkedIn | GitHub | Website

EDUCATION

Brown University, *Sc. B. Electrical Engineering*, 4.0/4.0 GPA — Providence, RI | **Expected Graduation May 2029**

Relevant Courses: Ordinary Differential Equations, Functional Programming, Statics & Dynamics, Data Structures & Algorithms

TECHNICAL SKILLS

Programming: C, Python (NumPy, Pandas, Matplotlib), Ada, MATLAB, Java, OCaml, Racket

Embedded Systems: Bare-metal firmware (RP2040, STM32), register-level programming, I2C/SPI/UART

Hardware: Schematic design(KiCad), soldering, CAD & CAM(SolidWorks, Fusion 360), FEA, 3D printing, sheet metal fabrication

Tools: Git/GitHub, CMake, OpenOCD, CMSIS-DAP, Saleae Logic, oscilloscope, serial monitor, Alire

RESEARCH

Interactive 3D Vision & Learning Lab (IVL), *Paid Researcher* — Providence, RI | June 2026 – Present

Advisor: Srinath Sridhar, Assistant Professor | Brown University

- Supporting multi-camera rig development: PCB schematic design, CAD modeling, and 3D-printed enclosure fabrication

INDUSTRY EXPERIENCE

AdaCore, *Technical Mentorship* — Providence, RI | December 2025 – Present

Mentor: Oliver Henley, GNAT Academic Coordinator | AdaCore

- Developed bare-metal Ada firmware for STM32 using register-level programming from ARM reference manuals

Persistence Labs, *Strategy Intern* — Charleston, SC | May 2024 – October 2025

- Facilitated beta testing with high school control groups, collected feedback across 10+ iterations to identify usability issues
- Synthesized testing results into reports and presented findings to the founder/CEO, guiding product design decisions
- Communicated user feedback to engineering team, directly informing feature prioritization and UI iteration

MIG3 Inc., *Part-Time Technical Assistant* — Irvine, CA | May 2022 – August 2025

- Provided technical support for office software and built Excel-based data tools, improving internal workflow efficiency

LEADERSHIP & ACTIVITIES

Brown Rocketry Club, *Avionics Team Lead, Co-President* — Providence, RI | September 2025 – Present

- Contributed to Brown Rocketry's first-ever IREC acceptance as sole avionics developer on the competing payload
- Designed IREC avionics electrical architecture and wiring diagrams for power distribution, sensor interfaces, and flight systems
- Performed FEA on avionics bay components and developed CAM-based manufacturing plans in Fusion 360
- Secured industry sponsorship by leading partner outreach and presenting club technical work to external stakeholders

Brown Space Engineering, *Antenna Team Member* — Providence, RI | March 2026 – Present

- Designing communications antennas (Nitinol dipole, copper patch), including specification and integration constraints
- Establishing feed architecture, impedance matching strategy, and EMC constraints on adjacent and connected hardware

AI & Robotics Ethics Society, *Research Team Member* — Providence, RI | September 2025 – Present

- Designing a survey and assessment to evaluate the effectiveness of Brown's Socially Responsible Computing curriculum
- Collaborating with the Computer Science Department and teaching assistants to collect and analyze student feedback

Project Amplify, *Founder & Director* — Irvine, CA | May 2024 – September 2025

- Founded a summer AI literacy program across 2+ school districts, designed and taught curriculum from scratch
- Taught students to train basic image recognition AI using Python, introducing coding and model evaluation with Pandas/NumPy

ENGINEERING PROJECTS

Bare-Metal Driver Stack, *Independent Project* — RP2040, C | May 2026 – Present

- Developed bare-metal GPIO, SysTick, UART, and I2C drivers for RP2040 in C using register-level programming without HAL
- Verified UART and I2C via logic analyzer and serial monitor; ISR-driven SysTick timing and atomic GPIO validated
- Designed layered driver structure separating register maps from driver logic for future Cortex-M portability

Rocket Payload Sensor System, *Sole Designer* — RP2040, Ada | February 2026 – April 2026

- Developed bare-metal Ada firmware for an RP2040 rocket payload; work mentored by AdaCore Engineering
- Deployed payload on a live high-power rocket launch; logged repeatable 12+ G flight data, validating system end to end
- Integrated BMP390 and LSM9DS1 sensors over I2C; logged 23+ hours of continuous flight and ground data
- Developed SPI flash logging pipeline for W25Q128 onboard storage; validated write/erase/read-back on hardware
- Debugged I2C addressing, SPI clocking, UART data loss, and power sequencing issues across embedded hardware and firmware

AWARDS & GRANTS

Brown UTRA Research Award, 2026 | **Brown Design Workshop Grant**, 2026 | **Taco Bell Ambition Accelerator Grant,** 2025